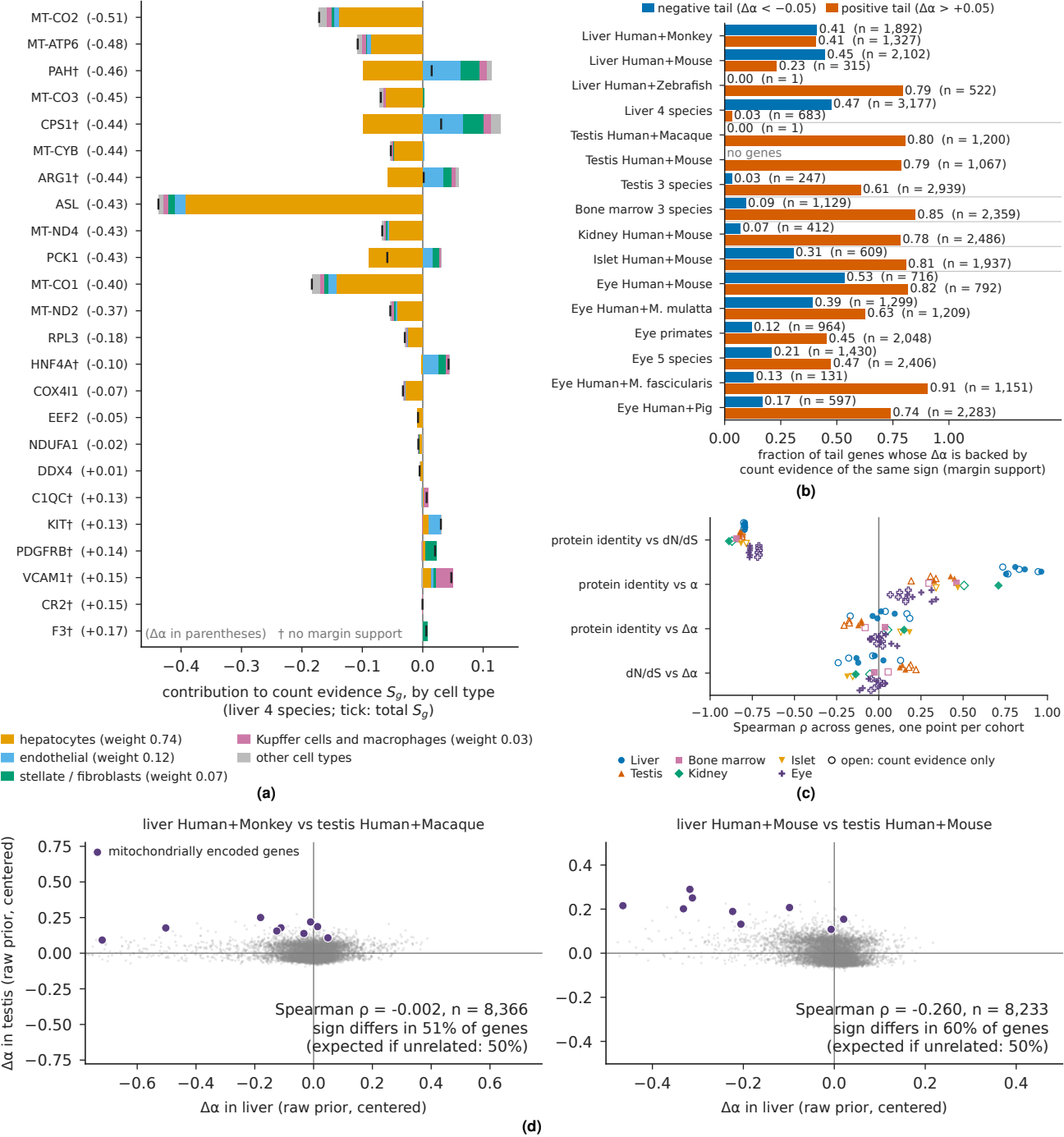

MT-CO2 (-0.51)
MT-ATP6 (-0.48)
PAH† (-0.46)
MT-CO3 (-0.45)
CPS1† (-0.44)
MT-CYB (-0.44)
ARG1† (-0.44)
ASL (-0.43)
MT-ND4 (-0.43)
PCK1 (-0.43)
MT-CO1 (-0.40)
MT-ND2 (-0.37)
RPL3 (-0.18)
HNF4A† (-0.10)
COX4I1 (-0.07)
EEF2 (-0.05)
NDUFA1 (-0.02)
DDX4 (+0.01)
C1QC† (+0.13)
KIT† (+0.13)
PDGFRB† (+0.14)
VCAM1† (+0.15)
CR2† (+0.15)
F3† (+0.17)
(Δα in parentheses) † no margin support
−0.4 −0.3 −0.2 −0.1 0.0 0.1
contribution to count evidence S_g, by cell type
(liver 4 species; tick: total S_g)
hepatocytes (weight 0.74)
endothelial (weight 0.12)
stellate / fibroblasts (weight 0.07)
Kupffer cells and macrophages (weight 0.03)
other cell types
(a)
negative tail (Δα < −0.05)
positive tail (Δα > +0.05)
Liver Human+Monkey
0.41 (n = 1,892)
0.41 (n = 1,327)
Liver Human+Mouse
0.45 (n = 2,102)
0.23 (n = 315)
Liver Human+Zebrafish
0.00 (n = 1)
0.79 (n = 522)
Liver 4 species
0.47 (n = 3,177)
0.03 (n = 683)
Testis Human+Macaque
0.00 (n = 1)
0.80 (n = 1,200)
Testis Human+Mouse
no genes
0.79 (n = 1,067)
Testis 3 species
0.03 (n = 247)
0.61 (n = 2,939)
Bone marrow 3 species
0.09 (n = 1,129)
0.85 (n = 2,359)
Kidney Human+Mouse
0.07 (n = 412)
0.78 (n = 2,486)
Islet Human+Mouse
0.31 (n = 609)
0.81 (n = 1,937)
Eye Human+Mouse
0.53 (n = 716)
0.82 (n = 792)
Eye Human+M. mulatta
0.39 (n = 1,299)
0.63 (n = 1,209)
Eye primates
0.12 (n = 964)
0.45 (n = 2,048)
Eye 5 species
0.21 (n = 1,430)
0.47 (n = 2,406)
Eye Human+M. fascicularis
0.13 (n = 131)
0.91 (n = 1,151)
Eye Human+Pig
0.17 (n = 597)
0.74 (n = 2,283)
0.00 0.25 0.50 0.75 1.00
fraction of tail genes whose Δα is backed by
count evidence of the same sign (margin support)
(b)
protein identity vs dN/dS
protein identity vs α
protein identity vs Δα
dN/dS vs Δα
−1.00 −0.75 −0.50 −0.25 0.00 0.25 0.50 0.75 1.00
Spearman ρ across genes, one point per cohort
Liver
Bone marrow
Islet
open: count evidence only
Testis
Kidney
Eye
(c)
liver Human+Monkey vs testis Human+Macaque
mitochondrially encoded genes
Spearman ρ = -0.002, n = 8,366
sign differs in 51% of genes
(expected if unrelated: 50%)
Δα in testis (raw prior, centered)
Δα in liver (raw prior, centered)
liver Human+Mouse vs testis Human+Mouse
Spearman ρ = -0.260, n = 8,233
sign differs in 60% of genes
(expected if unrelated: 50%)
Δα in liver (raw prior, centered)
(d)